ABCDEFGHIJKLMN
OPQRSTUVWXYZ
abcdefghijklmn
opqrstuvwxyz

The punchcutter begins by transferring the outline of a letter design to one end of a metal bar. The outer shape of the punch could be cut directly, but the internal curves of a small punch were particularly difficult as it was necessary to cut deep enough and straight into the metal. This was almost never done with cutting tools; a counterpunch, a type of punch used in the cutting of other punches, is used to create the negative space in or around a glyph. A counterpunch could be used to create this negative space, not just where the space was completely enclosed by the letter, but in any concavity, e.g. above and below the midbar in uppercase “H”. Of course, the counterpunch had to be harder than the punch itself. This was accomplished by heat tempering the counterpunch and softening the punch. Such a tool solved two issues, one technical and one aesthetic, that arose in punchcutting. Often the same counterpunch could be used for several letters in a typeface. For example, the negative space inside an uppercase “P” and “R” is usually very similar, and with the use of a counterpunch, they could be nearly identical. Counterpunches were regularly used in this way to give typefaces a more consistent look. The counterpunch would be struck into the face of the punch. The outer form of the letter is then shaped using files. To test the punch, the punchcutter makes an imprint on a piece of paper after the punch is heated on open flame. The soot left by the flame is used to create a sample on the paper, a smoke proof. Once the punches are read a mold could then be created from the punch by using the punch on a softer metal such as copper to create a matrix. Then, type metal, an alloy of lead, antimony, and tin, flows into the matrix to produce a single piece of type, ready for typesetting. One characteristic of type metal that makes it valuable for this use is that it expands as it cools, filling in any gaps present in the serifs and thinner portions of letters. This characteristic is shared by the bronze used to cast sculptures, but copper-based al loys generally have melting points that are too high to be convenient for typesetting. Water, silicon and bismuth are other substances with this property on freezing.